Capacity and Service-Level Stability in Quick Commerce Delivery Networks: A System Dynamics Study

Team Name: System Thinker

Team Members:
Ashish Shukla (25M1542)
Bhupesh Ranjan Jha (25M1541)
Motupalli Hemanth Kumar (25M1512)

Project Summary

Quick commerce platforms operate under strict 10-20 minute delivery commitments, creating intense pressure on capacity and fulfillment systems. Demand variability, rider availability, picking constraints, and inventory delays interact dynamically, often leading to backlog growth and unstable service levels.

This project develops a System Dynamics model to examine how these feedback effects influence delivery performance over time. The model captures order arrivals, backlog accumulation, capacity scaling, and service-level response within a representative quick commerce network.

Through simulation experiments, the study evaluates alternative capacity and response policies under demand uncertainty. The objective is to identify structural causes of instability and explore strategies that sustain reliable service without excessive overreaction in workforce or inventory decisions.

Detailed Project Description

Quick commerce has grown very rapidly in urban areas, especially with the rise of dark stores that allow deliveries within 10-20 minutes. While this model works well from a customer perspective, it creates operational challenges. Orders fluctuate during the day and spike during peak hours or promotions. At the same time, rider availability, picking capacity inside the dark store, and inventory replenishment cannot adjust immediately.

In such systems, small increases in demand can quickly lead to backlog accumulation. As orders pile up, delivery time increases and customer satisfaction may decline. Managers respond by hiring more riders, extending shifts, or increasing stock levels. However, these adjustments involve delays. Hiring takes time, onboarding takes time, and inventory replenishment depends on supplier lead times. Because of these delays, the system may overreact or underreact, leading to oscillations in backlog, capacity, and service level.

Research in industrial dynamics and supply chain systems shows that delays combined with feedback often generate instability (Forrester, 1961; Sterman, 2000). In last-mile delivery systems, short delivery windows increase sensitivity to demand fluctuations (Boysen et al., 2021). Urban logistics studies also highlight that operational efficiency depends on how capacity is adjusted over time rather than only increasing resources (Mangiaracina et al., 2019). These findings support the use of System Dynamics to understand structural behavior in quick commerce networks.

For this project, we will build a Stock-and-Flow Diagram (SFD) in Vensim to represent a simplified quick commerce dark store system. The model will focus on three main stocks: order backlog, rider workforce, and inventory level. Backlog increases when order arrival exceeds fulfillment. Rider workforce changes through hiring and attrition. Inventory increases through replenishment and decreases through sales. Order fulfillment will depend on the minimum of available inventory, picking capacity, and rider delivery capacity.

We will include realistic delay structures such as hiring adjustment time and replenishment lead time. Delivery delay will be modeled as a function of backlog relative to rider capacity. If backlog becomes too large relative to available riders, delivery delay increases significantly. This delay can influence managerial response intensity, creating feedback between service performance and capacity decisions.

The model is expected to capture three main feedback patterns: a reinforcing demand-pressure loop, a balancing capacity-adjustment loop, and an inventory-control loop. The interaction of these loops, especially under demand surge conditions, may create oscillatory behavior similar to classical industrial dynamics models.

We plan to simulate the system over several months and test policies such as faster versus slower hiring responses, different safety inventory levels, gradual capacity scaling, and demand surge scenarios. We will also observe how strict service commitments influence system stability. Performance will be evaluated using average delivery delay, backlog size, rider utilization, inventory stability, and overall responsiveness of the system.

By analyzing simulation results, we aim to identify leverage points that improve stability without excessive cost escalation. The overall goal is to provide a structured understanding of how capacity and inventory decisions affect service-level stability in quick commerce delivery networks.

Initial References

1. Forrester, J. W. (1961). *Industrial Dynamics*. MIT Press.
2. Sterman, J. D. (2000). *Business Dynamics: Systems Thinking and Modeling for a Complex World*. McGraw-Hill.
3. Boysen, N., Fedtke, S., & Schwerdfeger, S. (2021). Last-mile delivery concepts: A survey from an operational research perspective. *European Journal of Operational Research*, 289(2), 343-361.
4. Mangiaracina, R., Perego, A., & Tumino, A. (2019). Innovative solutions to increase last-mile delivery efficiency in urban areas. *Sustainability*, 11(9).